

Comparative Analysis of NOMA and RSMA in Downlink Networks Under Nakagami-m Fading

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Abstract

This paper presents a comparative analysis of Non-Orthogonal Multiple Access (NOMA) and Rate-Splitting Multiple Access (RSMA) for downlink networks operating under Nakagami-m fading. To comprehensively evaluate their capabilities, we investigate two distinct communication scenarios. The first scenario examines a downlink Integrated Sensing and Communication (ISAC) system operating with imperfect Successive Interference Cancellation (SIC). In this environment, we assess Outage Probability (OP), Ergodic Rate (ER), and the Probability of Detection (PoD) for sensing targets. Simulations reveal a fundamental trade-off: RSMA delivers superior OP for far users and better target detection (PoD), whereas NOMA achieves higher overall Ergodic Rates and better outage reliability for near users. The second scenario explores a three-node cooperative NOMA architecture utilizing Decode-and-Forward (DF) relaying to compare Full-Duplex (FD) and Half-Duplex (HD) transmission modes. We analyze the severe impact of self-loop interference on delay-limited system throughput and sum rates. The results demonstrate that while FD-NOMA maximizes throughput at low-to-medium Signal-to-Noise Ratios (SNRs), it is bottlenecked by an interference-induced error floor at higher SNRs. Conversely, HD-NOMA overcomes this limitation, showcasing higher peak throughput and reliability in high-power regimes. Ultimately, this study highlights the critical trade-offs between user fairness, capacity ceilings, and interference mitigation when deploying advanced multiple access schemes in next-generation networks.

I. INTRODUCTION

A. *Brief introduction to your research topic*

The explosive growth of wireless data traffic and the stringent demands of next-generation (6G) networks require communication systems that can deliver massive connectivity, ultra-high spectral efficiency, and versatile functionalities. To meet these demands, advanced multiple access schemes have become a critical focal point of wireless research. Non-Orthogonal Multiple Access (NOMA) has established itself as a leading technology by allowing multiple users to share the same time and frequency resources, separating them in the power domain and utilizing Successive Interference Cancellation (SIC) at the receiver. Recently, Rate-Splitting Multiple Access (RSMA)

has emerged as a powerful, generalized alternative. By splitting user messages into common and private streams, RSMA flexibly manages interference and often overcomes the strict power-allocation constraints that limit traditional NOMA systems.

Beyond connectivity, future networks are expected to merge communication and radar functionalities through Integrated Sensing and Communication (ISAC). Simultaneously, expanding network coverage and reliability remains paramount, driving the development of Cooperative Relaying techniques where users assist one another to forward data. However, implementing these advanced frameworks introduces severe practical challenges, such as channel estimation errors, imperfect SIC, and self-loop interference in full-duplex systems. This paper aims to provide a comparative performance analysis of NOMA and RSMA under these practical constraints in Nakagami-m fading environments, specifically evaluating their efficacy in dual-scenario setups involving ISAC target detection and cooperative decode-and-forward relaying.

B. Related works

This study builds directly upon the foundational analyses provided by two primary research papers investigating advanced multiple access techniques under practical channel constraints.

The first paper, "Multiple Access Enabled Integrated Sensing and Communication With Imperfect SIC: Non-orthogonal vs. Rate Splitting," explores the integration of sensing and communication (ISAC) functionalities. The authors provide a rigorous mathematical framework comparing NOMA and RSMA in downlink ISAC systems. A key contribution of their work is the realistic inclusion of Channel Estimation Errors (CEEs) and imperfect SIC, demonstrating how these physical layer impairments degrade performance. They establish that while NOMA provides higher Ergodic Rates (ER), RSMA achieves superior Outage Probability (OP) for far users and better Probability of Detection (PoD) for sensing targets.

The second paper, "Performance Analysis of Downlink Cooperative User Relaying FD/HD NOMA Over Nakagami-m Fading Channels," focuses on coverage extension through cooperative communications. The authors analyze a three-node NOMA network where a near user acts as a Decode-and-Forward (DF) relay for a far user. Their key contribution is a comprehensive comparison between Full-Duplex (FD) and Half-Duplex (HD) relaying modes. The study critically highlights the bottleneck of self-loop interference in FD-NOMA, showing that while FD provides high throughput at low-to-medium SNRs, it suffers from an error floor at high SNRs, a limitation that HD-NOMA reliably overcomes.

Together, these works provide the necessary mathematical models—from outage probabilities to ergodic capacities—that form the foundation of the comparative scenarios simulated and evaluated in this paper.

II. PROOF OF CONCEPT

NOMA vs RSMA in Downlink ISAC with Imperfect SIC

A. System model

1) *NOMA ISAC System*: For communication, the BS transmits signals $\mathbf{x}_S^{NO}$ to communicate with the near user D_n and the far user D_f , as well as sense targets, which is expressed as,

$$\mathbf{x}_S^{NO} = \sqrt{a_n P_S} \mathbf{w}_n x_n + \sqrt{a_f P_S} \mathbf{w}_f x_f + \sqrt{a_r P_S} \mathbf{w}_r x_r \quad (1)$$

where P_S denotes the transmit power of S , x_n and x_f represent the expected signals for D_n and D_f respectively, while x_r is the sensing signal with $\mathbb{E}[|x_n|^2] = \mathbb{E}[|x_f|^2] = \mathbb{E}[|x_r|^2] = 1$. a_n and a_f are the power allocation coefficients for D_n and D_f respectively, and a_r is the power allocation coefficient for sensing with $a_r > a_f > a_n$ and $a_r + a_f + a_n = 1$. Furthermore, $\mathbf{w}_n \in \mathbb{C}^{M \times 1}$, $\mathbf{w}_f \in \mathbb{C}^{M \times 1}$ and $\mathbf{w}_r \in \mathbb{C}^{M \times 1}$ denote the transmit beamforming (BF) vectors for x_n , x_f and x_r respectively, with $\|\mathbf{w}_n\|^2 = \|\mathbf{w}_f\|^2 = \|\mathbf{w}_r\|^2 = 1$. Furthermore, by accounting for CEEs, the channel coefficients of S to D_n and S to D_f are defined as $\mathbf{h}_{SD_n} = \hat{\mathbf{h}}_{SD_n}^H + \mathbf{e}_{SD_n}$ and $\mathbf{h}_{SD_f} = \hat{\mathbf{h}}_{SD_f}^H + \mathbf{e}_{SD_f}$ respectively, where $\hat{\mathbf{h}}_{SD_n} = [\hat{h}_{SD_n^1}, \hat{h}_{SD_n^2}, \dots, \hat{h}_{SD_n^M}]^T \in \mathbb{C}^{M \times 1}$ and $\hat{\mathbf{h}}_{SD_f} = [\hat{h}_{SD_f^1}, \hat{h}_{SD_f^2}, \dots, \hat{h}_{SD_f^M}]^T \in \mathbb{C}^{M \times 1}$ are the estimated communication channel coefficients with $\hat{h}_{SD_n^m}$ and $\hat{h}_{SD_f^m}$ $m \in [1, M]$ denoting channels between the m -th transmit antenna of the BS and the users, $\mathbf{e}_{SD_n} \sim \mathcal{CN}(0, \sigma_{SD_n}^2 \mathbf{I}) \in \mathbb{C}^{M \times 1}$ and $\mathbf{e}_{SD_f} \sim \mathcal{CN}(0, \sigma_{SD_f}^2 \mathbf{I}) \in \mathbb{C}^{M \times 1}$ denote the corresponding CEEs. Therefore the received signals at D_n and D_f for the NOMA ISAC system considered are respectively expressed as

$$y_n^{NO} = \mathbf{h}_{SD_n} \mathbf{x}_S^{NO} + n_{D_n}, \quad (2)$$

$$y_f^{NO} = \mathbf{h}_{SD_f} \mathbf{x}_S^{NO} + n_{D_f} \quad (3)$$

where $n_{D_n} \sim \mathcal{CN}(0, N_0)$ and $n_{D_f} \sim \mathcal{CN}(0, N_0)$ denote the AWGN at D_n and D_f respectively.

According to NOMA protocol, by considering ipSIC, the signals x_r , x_f and x_n are first detected and then removed at D_n with received SNRs as,

$$\gamma_{SD_n}^{NO, x_r} = \frac{\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r\|^2 a_r \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_n\|^2 a_n + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_f\|^2 a_f + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (4)$$

$$\gamma_{SD_n}^{NO,x_f} = \frac{\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_f\|^2 a_f \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_n\|^2 a_n + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r\|^2 \epsilon a_f + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (5)$$

$$\gamma_{SD_n}^{NO,x_n} = \frac{\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_n\|^2 a_n \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_f\|^2 a_f + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r\|^2 \epsilon a_r + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (6)$$

where we have $\gamma_S = \frac{P_S}{N_0}$ and $\epsilon \in [0, 1]$ denotes the ipSIC parameter. Peculiarly, $\epsilon = 0$ denotes perfect SIC and $\epsilon = 1$ denotes the case of completely failed SIC.

Then, the signals x_r and x_f are first detected and then removed at D_f , with received SNRs as,

$$\gamma_{SD_f}^{NO,x_r} = \frac{\|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_r\|^2 a_r \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_n\|^2 a_n + \|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_f\|^2 a_f + \sigma_{SD_f}^2 \right) \gamma_S + 1} \quad (7)$$

$$\gamma_{SD_f}^{NO,x_f} = \frac{\|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_f\|^2 a_f \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_n\|^2 a_n + \|\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_r\|^2 \epsilon a_r + \sigma_{SD_f}^2 \right) \gamma_S + 1} \quad (8)$$

As for sensing, the BS transmits $\mathbf{x}_S^{NO}$, which upon encountering the target, is reflected back as the echo signal. The received signals at the BS can be divided into three parts: (1) The target echo; (2) The communication channel interference reflected by the target; and (3) AWGN. Therefore, the received echo for NOMA ISAC system is,

$$\mathbf{y}_S^{NO} = \underbrace{\mathbf{G} \left(\sqrt{a_r P_S} \mathbf{w}_r x_r + \sqrt{a_f P_S} \mathbf{w}_f x_f \right)}_{\text{Sensing Echo}} + \underbrace{\mathbf{G} \left(\sqrt{a_n P_S} \mathbf{w}_n s_n \right)}_{\text{Interference}} + \underbrace{\mathbf{n}_S}_{\text{AWGN}} \quad (9)$$

where $\mathbf{n}_S \sim \mathcal{CN}(0, N_0 \mathbf{I}) \in \mathbb{C}^{N \times 1}$ denotes the AWGN at the BS. $\mathbf{G} = \sum_{k \in K} \sqrt{\beta_k} \mathbf{a}(\theta_k) \mathbf{b}^T(\theta_k)$ characterises the response matrix of K targets, where β_k is the attenuation of the echo signal for the k th target, θ_k is the direction of arrival of the k th target, $\mathbf{a}(\theta_k) \in \mathbb{C}^{N \times 1}$ and $\mathbf{b}(\theta_k) \in \mathbb{C}^{M \times 1}$ are the received and transmitted array steering vectors respectively associated with the k th target, which are expressed as,

$$\mathbf{a}(\theta_k) = \begin{bmatrix} 1 & e^{j \frac{2\pi d}{\lambda} \sin \theta_k} & \dots & e^{j \frac{2\pi (N-1)d}{\lambda} \sin \theta_k} \end{bmatrix} \quad (10)$$

$$\mathbf{b}(\theta_k) = \begin{bmatrix} 1 & e^{j \frac{2\pi d}{\lambda} \sin \theta_k} & \dots & e^{j \frac{2\pi (M-1)d}{\lambda} \sin \theta_k} \end{bmatrix} \quad (11)$$

where d and λ are the antenna spacing and wavelength respectively.

2) *RSMA ISAC System*: In RSMA ISAC systems, communication and sensing signals are decomposed and reorganised into one common and multiple private signals. Thus, the transmitted signals can be written as,

$$\mathbf{x}_S^{NO} = \underbrace{\sqrt{a_{n,p}}P_S\mathbf{w}_{n,p}x_{n,p} + \sqrt{a_{f,p}}P_S\mathbf{w}_{f,p}x_{f,p}}_{x_p} + \underbrace{\sqrt{a_{S,c}}P_S\mathbf{w}_{S,c}x_{S,c}}_{x_c} + \sqrt{a_r}P_S\mathbf{w}_r x_r \quad (12)$$

where $x_{S,c}$ is the common signal, $x_{n,p}$ and $x_{f,p}$ are the private signals for D_n and D_f respectively with $\mathbb{E}[|x_{S,c}|^2] = \mathbb{E}[|x_{n,p}|^2] = \mathbb{E}[|x_{f,p}|^2] = 1$. While a_c , $a_{n,p}$ and $a_{f,p}$ are the power allocation parameters assigned to the transmitted signals with $a_c + a_{n,p} + a_{f,p} + a_r = 1$. Furthermore, $\mathbf{w}_{S,c} \in \mathbb{C}^{M \times 1}$, $\mathbf{w}_{n,p} \in \mathbb{C}^{M \times 1}$ and $\mathbf{w}_{f,p} \in \mathbb{C}^{M \times 1}$ are the transmit BF vectors of the common and private signals for D_n and D_f with $\|\mathbf{w}_{S,c}\|^2 = \|\mathbf{w}_{n,p}\|^2 = \|\mathbf{w}_{f,p}\|^2 = 1$, respectively. Specifically, the common message $x_{S,c}$ is constructed from parts of both users' original messages. The original messages $\mathbf{m}_n$ and $\mathbf{m}_f$ are split into common and private parts i.e. $\mathbf{m}_n \rightarrow \{\mathbf{m}_{n,c}, \mathbf{m}_{n,p}\}$ and $\mathbf{m}_f \rightarrow \{\mathbf{m}_{f,c}, \mathbf{m}_{f,p}\}$. The common message is then formed as $\mathbf{m}_{S,c} = \{\mathbf{w}_{n,c}, \mathbf{w}_{f,c}\}$, which is jointly encoded as $x_{S,c}$. As a further step, the common rate R_c^{RS} is correspondingly decomposed into two components $R_c^{RS} = R_{D_n,c}^{RS} + R_{D_f,c}^{RS}$, where $R_{D_n,c}^{RS}$ denotes the rate allocated to transmit D_n 's common signal and $R_{D_f,c}^{RS}$ denotes the rate allocated to transmit D_f 's common signal. Based on (12), the received signal at D_n and D_f for the RSMA ISAC system considered can be expressed as,

$$y_n^{RS} = \mathbf{h}_{SD_n} \mathbf{x}_S^{RS} + n_{D_n} \quad (13)$$

$$y_f^{RS} = \mathbf{h}_{SD_f} \mathbf{x}_S^{RS} + n_{D_f} \quad (14)$$

According to the RSMA protocol, the sensing signal, common signal and private signals are decoded in sequence by employing SIC at the users. By considering ipSIC, the received x_r , $x_{S,c}$ and $x_{n,p}$ as well as x_r , $x_{S,c}$ and $x_{f,p}$ are first detected and then removed at D_n and D_f respectively, with received SNRs,

$$\gamma_{SD_n}^{RS,x_r} = \frac{\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r\|^2 a_r \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{n,p}\|^2 a_{n,p} + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{f,p}\|^2 a_{f,p} + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{S,c}\|^2 a_c + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (15)$$

$$\gamma_{SD_n}^{RS,x_c} = \frac{\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{S,c}\|^2 a_c \gamma_S}{\left(\|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{n,p}\|^2 a_{n,p} + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{f,p}\|^2 a_{f,p} + \|\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r\|^2 a_r + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (16)$$

$$\gamma_{SD_n}^{RS,x_{n,p}} = \frac{||\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{n,p}||^2 a_{n,p} \gamma_S}{\left(||\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_c||^2 a_c + ||\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_{f,p}||^2 a_{f,p} + ||\hat{\mathbf{h}}_{SD_n}^H \mathbf{w}_r||^2 \epsilon a_r + \sigma_{SD_n}^2 \right) \gamma_S + 1} \quad (17)$$

$$\gamma_{SD_f}^{RS,x_r} = \frac{||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_r||^2 a_r \gamma_S}{\left(||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{n,p}||^2 a_{n,p} + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{f,p}||^2 a_{f,p} + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{S,c}||^2 a_c + \sigma_{SD_f}^2 \right) \gamma_S + 1} \quad (18)$$

$$\gamma_{SD_f}^{RS,x_c} = \frac{||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{S,c}||^2 a_c \gamma_S}{\left(||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{n,p}||^2 a_{n,p} + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{f,p}||^2 a_{f,p} + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_r||^2 \epsilon a_r + \sigma_{SD_f}^2 \right) \gamma_S + 1} \quad (19)$$

$$\gamma_{SD_f}^{RS,x_{n,p}} = \frac{||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{n,p}||^2 a_{n,p} \gamma_S}{\left(||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_c||^2 a_c + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_{f,p}||^2 a_{f,p} + ||\hat{\mathbf{h}}_{SD_f}^H \mathbf{w}_r||^2 \epsilon a_r + \sigma_{SD_f}^2 \right) \gamma_S + 1} \quad (20)$$

As for the sensing, the BS transmits $\mathbf{x}_S^{RS}$ to detect the target, in which x_r is deemed as sensing signal. Similar to (9), the received echo for the RSMA ISAC system considered can be expressed as,

$$\mathbf{y}_S^{RS} = \underbrace{\mathbf{G} \left(x_c + \sqrt{a_r P_S} \mathbf{w}_r x_r \right)}_{\text{Sensing Echo}} + \underbrace{\mathbf{G} x_p}_{\text{Interference}} + \underbrace{\mathbf{n}_S}_{\text{AWGN}} \quad (21)$$

B. Analysis

OP Analysis:

1) *NOMA ISAC System:* An outage event is occurred at D_f , if either x_r or x_f cannot be decoded successfully. Thus the OP of D_f for the NOMA ISAC system is given as,

$$\mathbf{P}_{D_f}^{NO} = 1 - \Pr \left(\gamma_{SD_f}^{NO,x_r} > \gamma_{\text{thr}}, \gamma_{SD_f}^{NO,x_f} > \gamma_{\text{thf}} \right) \quad (22)$$

where γ_{thr} and γ_{thf} are SNR thresholds for x_r and x_f respectively. The exact OP expression for D_f for the NOMA ISAC system is derived by the authors of the original paper as,

$$\mathbf{P}_{D_f}^{NO} = \frac{\gamma (m_{SD_f} M, m_{SD_f} \phi' / \Omega_{SD_f})}{\Gamma (m_S D_f M)} \quad (23)$$

where $\phi' = \max \{ \phi'_1, \phi'_2 \}$, with

$$\phi'_1 = \frac{\left(\sigma_{SD_f}^2 + 1 \right) \gamma_{\text{thr}}}{a_r \gamma_S - (a_n + a_f) \gamma_S \gamma_{\text{thr}}}$$

$$\phi'_2 = \frac{\left(\sigma_{SD_f}^2 + 1 \right) \gamma_{\text{thf}}}{a_f \gamma_S - (\epsilon a_r + a_n) \gamma_S \gamma_{\text{thf}}}$$

with $a_r > (a_n + a_f) \gamma_{\text{thr}}$ and $a_f > (\epsilon a_r + a_n) \gamma_{\text{thf}}$. Otherwise $\mathbf{P}_{D_f}^{NO} = 1$.

An outage event is occurred at D_n , if one of x_r , x_f and x_n cannot be decoded successfully. Thus the OP of D_n for the NOMA ISAC system is given as,

$$\mathbf{P}_{D_f}^{NO} = 1 - \Pr \left(\gamma_{SD_n}^{NO, x_r} > \gamma_{\text{thr}}, \gamma_{SD_n}^{NO, x_f} > \gamma_{\text{thf}}, \gamma_{SD_n}^{NO, x_n} > \gamma_{\text{thn}} \right) \quad (24)$$

where γ_{thn} is the SNR threshold of x_n . The exact OP expression for D_n for the NOMA ISAC system is given as,

$$\mathbf{P}_{D_n}^{NO} = \frac{\gamma (m_{SD_n} M, m_{SD_n} \varphi' / \Omega_{SD_n})}{\Gamma (m_S D_n M)} \quad (25)$$

where $\varphi' = \max \{\phi', \varphi'_3\}$, and further φ'_3 is given as

$$\varphi'_3 = \frac{(\sigma_{SD_n}^2 + 1) \gamma_{\text{thn}} + 1}{a_n \gamma_S - \epsilon(a_f + a_r) \gamma_S \gamma_{\text{thn}}}$$

with $a_n > \epsilon(a_f + a_r) \gamma_{\text{thn}}$. Otherwise, $\mathbf{P}_{D_n}^{NO} = 1$.

2) *RSMA ISAC System*: An outage event is occurred at D_f , if one of x_r , $x_{S,c}$ and $x_{f,p}$ cannot be decoded successfully. Thus the OP of D_f for the RSMA ISAC system is given as,

$$\mathbf{P}_{D_f}^{RS} = 1 - \Pr \left(\gamma_{SD_f}^{RS, x_r} > \gamma_{\text{thr}}, \gamma_{SD_f}^{RS, x_c} > \gamma_{\text{thc}}, \gamma_{SD_f}^{RS, x_{f,p}} > \gamma_{\text{thf,p}} \right) \quad (26)$$

where γ_{thc} and $\gamma_{\text{thf,p}}$ denote the SNR thresholds of $x_{S,c}$ and $x_{f,p}$ respectively. The exact OP expression of D_f for the RSMA ISAC system is given as,

$$\mathbf{P}_{D_f}^{RS} = \frac{\gamma (m_{SD_f} M, m_{SD_f} \phi / \Omega_{SD_f})}{\Gamma (m_S D_f M)} \quad (27)$$

where $\phi = \max \{\phi_1, \phi_2, \varphi_1\}$, with

$$\begin{aligned} \phi_1 &= \frac{(\sigma_{SD_f}^2 + 1) \gamma_{\text{thf,p}}}{a_{f,p} \gamma_S - [a_{n,p} + \epsilon(a_c + a_r)] \gamma_S \gamma_{\text{thf,p}}} \\ \phi_2 &= \frac{(\sigma_{SD_f}^2 + 1) \gamma_{\text{thr}}}{a_r \gamma_S - (a_c + a_{n,p} + a_{f,p}) \gamma_S \gamma_{\text{thr}}} \\ \varphi_1 &= \frac{(\sigma_{SD_f}^2 + 1) \gamma_{\text{thc}}}{a_c \gamma_S - (a_{f,p} + a_{n,p} + \epsilon a_r) \gamma_S \gamma_{\text{thc}}} \end{aligned}$$

with $a_{f,p} > [a_{n,p} + \epsilon(a_c + a_r)] \gamma_{\text{thf,p}}$, $a_r > (a_c + a_{n,p} + a_{f,p}) \gamma_{\text{thr}}$ and $a_c > (a_{f,p} + a_{n,p} + \epsilon a_r) \gamma_{\text{thc}}$. Otherwise $\mathbf{P}_{D_f}^{RS} = 1$.

An outage event is occurred at D_n , if one of x_r , $x_{S,c}$ and $x_{n,p}$ cannot be decoded successfully. Thus the OP of D_n for the RSMA ISAC system is given as,

$$\mathbf{P}_{D_n}^{RS} = 1 - \Pr \left(\gamma_{SD_f}^{RS, x_r} > \gamma_{\text{thr}}, \gamma_{SD_f}^{RS, x_c} > \gamma_{\text{thc}}, \gamma_{SD_f}^{RS, x_{n,p}} > \gamma_{\text{thn,p}} \right) \quad (28)$$

where $\gamma_{\text{thn,p}}$ denotes the SNR threshold for $x_{n,p}$. The exact OP expression of D_n for the RSMA ISAC system is expressed as,

$$\mathbf{P}_{D_n}^{RS} = \frac{\gamma (m_{SD_n} M, m_{SD_f} \varphi / \Omega_{SD_n})}{\Gamma (m_S D_n M)} \quad (29)$$

where $\varphi = \max \{\varphi_1, \varphi_2, \phi_2\}$ and φ_2 is further given by,

$$\varphi_2 = \frac{(\sigma_{SD_n}^2 + 1) \gamma_{\text{thn,p}}}{a_{n,p} \gamma_S - [a_{f,p} + \epsilon(a_c + a_r)] \gamma_S \gamma_{\text{thn,p}}}$$

with $a_{n,p} > [a_{f,p} + \epsilon(a_c + a_r)] \gamma_S \gamma_{\text{thn,p}}$. Otherwise $\mathbf{P}_{D_n}^{RS} = 1$.

ER Analysis:

1) *NOMA ISAC System:* The achievable rates for D_f and D_n in the NOMA ISAC system can be respectively expressed as,

$$R_{D_f}^{NO} = \log_2 \left(1 + \min \left(\gamma_{SD_n}^{NO, x_f}, \gamma_{SD_f}^{NO, x_f} \right) \right) \quad (30)$$

$$R_{D_n}^{NO} = \log_2 \left(1 + \gamma_{SD_n}^{NO, x_n} \right) \quad (31)$$

The ERs for D_f and D_n for the NOMA ISAC system is respectively then,

$$R_{D_f,ave}^{NO} = \mathbb{E} \left[R_{D_f}^{NO} \right] \quad (32)$$

$$R_{D_n,ave}^{NO} = \mathbb{E} \left[R_{D_n}^{NO} \right] \quad (33)$$

The asymptotic ER expression of D_f and D_n in high SNR region for the NOMA ISAC system is,

$$R_{D_f,ave}^{NO,h} = \log_2 \left(1 + \frac{a_f}{a_n + \epsilon a_r + \sigma_{SD_f}^2} \right) \quad (34)$$

$$R_{D_n,ave}^{NO,h} = \log_2 \left(1 + \frac{a_n}{\epsilon(a_r + a_f) + \sigma_{SD_n}^2} \right) \quad (35)$$

2) *RSMA ISAC System:* The effective common rate for the user is defined as,

$$R_c^{RS} = \min \left\{ \log_2 \left(1 + \gamma_{SD_n}^{RS, x_c} \right), \log_2 \left(1 + \gamma_{SD_f}^{RS, x_c} \right) \right\} \quad (36)$$

Consequently, the achievable rates for D_f and D_n in the RSMA ISAC system can be expressed as,

$$R_{D_n}^{RS} = R_{D_n,c}^{RS} + \log_2 \left(1 + \gamma_{SD_n}^{RS, x_{n,p}} \right) \quad (37)$$

$$R_{D_f}^{RS} = R_{D_f,c}^{RS} + \log_2 \left(1 + \gamma_{SD_f}^{RS, x_{f,p}} \right) \quad (38)$$

For the purpose of a fair baseline comparison, we assume equal split of the common rate $R_{D_n,c}^{RS} = R_{D_f,c}^{RS} = \frac{R_c^{RS}}{2}$.

The ERs for D_f and D_n for the RSMA ISAC system is respectively then,

$$R_{D_f,ave}^{RS} = \mathbb{E} [R_{D_f}^{RS}] \quad (39)$$

$$R_{D_n,ave}^{RS} = \mathbb{E} [R_{D_n}^{RS}] \quad (40)$$

The asymptotic ER expression of D_f and D_n in high SNR region for the RSMA ISAC system are given as,

$$R_{D_f,ave}^{RS,h} = \log_2 \left(1 + \frac{a_c}{a_{n,p} + a_{f,p} + \epsilon a_r + \sigma_{SD_f}^2} \right) + \log_2 \left(1 + \frac{a_{n,p}}{\epsilon(a_c + a_r) + a_{f,p} + \sigma_{SD_f}^2} \right) \quad (41)$$

$$R_{D_n,ave}^{RS,h} = \log_2 \left(1 + \frac{a_c}{a_{n,p} + a_{f,p} + \epsilon a_r + \sigma_{SD_n}^2} \right) + \log_2 \left(1 + \frac{a_{n,p}}{\epsilon(a_c + a_r) + a_{n,p} + \sigma_{SD_n}^2} \right) \quad (42)$$

PoD Analysis:

1) *NOMA ISAC System:* The alternative and null hypotheses for the NOMA ISAC system can be written as,

$$\mathcal{H}_0 : \mathbf{y}_S^{NO} = \mathbf{G} \sqrt{a_n P_S} \mathbf{w}_f x_f + \mathbf{G}_{k \in K \setminus t} \left(\sqrt{a_f P_S} \mathbf{w}_n x_n + \sqrt{a_r P_S} \mathbf{w}_r x_r \right) + \mathbf{n}_S \quad (43)$$

$$\mathcal{H}_1 : \mathbf{y}_S^{NO} = \mathbf{G} \left(\sqrt{a_f P_S} \mathbf{w}_n x_n + \sqrt{a_f P_S} \mathbf{w}_f x_f \right) + \sqrt{a_r P_S} \mathbf{w}_r x_r + \mathbf{n}_S \quad (44)$$

From this the PoFA $\mathbf{P}_{fa}^{NO,t}$ is expressed as,

$$\mathbf{P}_{fa}^{NO,t} = \mathcal{Q}_3 \left(\sqrt{\frac{\text{tr} \left(P_S \left(a_n \sum_{k \in K} \mathbf{G} \mathbf{G}^H + (a_f + a_r) \sum_{k \in K \setminus t} \mathbf{G} \mathbf{G}^H \right) \right)}{N_0}}, \sqrt{\frac{2\xi}{N_0}} \right) \quad (45)$$

Furthermore, the PoD $\mathbf{P}_d^{NO,t}$ is expressed as,

$$\mathbf{P}_d^{NO,t} = \mathcal{Q}_3 \left(\sqrt{\frac{\text{tr} \left(P_S \sum_{k \in K} \mathbf{G} \mathbf{G}^H \right)}{N_0}}, \sqrt{\frac{2\xi}{N_0}} \right) \quad (46)$$

2) *RSMA ISAC System:* The alternative and null hypotheses for the RSMA ISAC system can be written as,

$$\mathcal{H}_0 : \mathbf{y}_S^{RS} = \mathbf{G} \left(\sqrt{a_{n,p} P_S} \mathbf{w}_{n,p} x_{n,p} + \sqrt{a_{f,p} P_S} \mathbf{w}_{f,p} x_{f,p} \right) + \mathbf{G}_{k \in K \setminus t} \left(x_c + \sqrt{a_r P_S} \mathbf{w}_r x_r \right) + \mathbf{n}_S \quad (47)$$

$$\mathcal{H}_1 : \mathbf{y}_S^{NO} = \mathbf{G} \left(x_c + x_p + \sqrt{a_r P_S} \mathbf{w}_r x_r \right) + \mathbf{n}_S \quad (48)$$

From this the PoFA $\mathbf{P}_{fa}^{RS,t}$ is expressed as,

$$\mathbf{P}_{fa}^{RS,t} = \mathcal{Q}_3 \left(\sqrt{\frac{\text{tr} \left((a_{n,p} + a_{f,p}) P_S \sum_{k \in K} \mathbf{G} \mathbf{G}^H + (a_c + a_r) P_S \sum_{k \in K \setminus t} \mathbf{G} \mathbf{G}^H \right)}{N_0}}, \sqrt{\frac{2\xi}{N_0}} \right) \quad (49)$$

Furthermore, the PoD $\mathbf{P}_d^{RS,t}$ is expressed as,

$$\mathbf{P}_d^{RS,t} = \mathcal{Q}_3 \left(\sqrt{\frac{\text{tr} (P_S \sum_{k \in K} \mathbf{G} \mathbf{G}^H)}{N_0}}, \sqrt{\frac{2\xi}{N_0}} \right) \quad (50)$$

C. Results and Discussions

Simulations were performed in MATLAB to validate the above discussions. The following values of parameters were used,

TABLE I
SUMMARY OF THE NOTATIONS

Parameter	Value	Parameter	Value	Parameter	Value
m_{SD_n}	2	m_{SD_f}	1	Ω_{SD_n}	1
Ω_{SD_f}	1	γ_{thr}	0.5	γ_{thn}	0.6
γ_{thf}	0.8	γ_{thc}	0.6	$\gamma_{thn,p}$	0.26
$\gamma_{thf,p}$	0.42	ϵ	0.01	N_0	1
a_c	0.3	$a_{n,p}$	0.05	$a_{f,p}$	0.15
a_r	0.5	a_n	0.2	a_f	0.3
λ	0.01	ρ, G	1	K	2

Fig. 1 illustrates the OP vs SNR for both NOMA and RSMA schemes. The theoretical derived expressions closely follow the Monte-Carlo simulations. CEEs parameter introduce an error floor to the OP in high SNR region. RSMA offers superior OP performance compared to NOMA for the far user but RSMA is better for near user as can be seen from the plot.

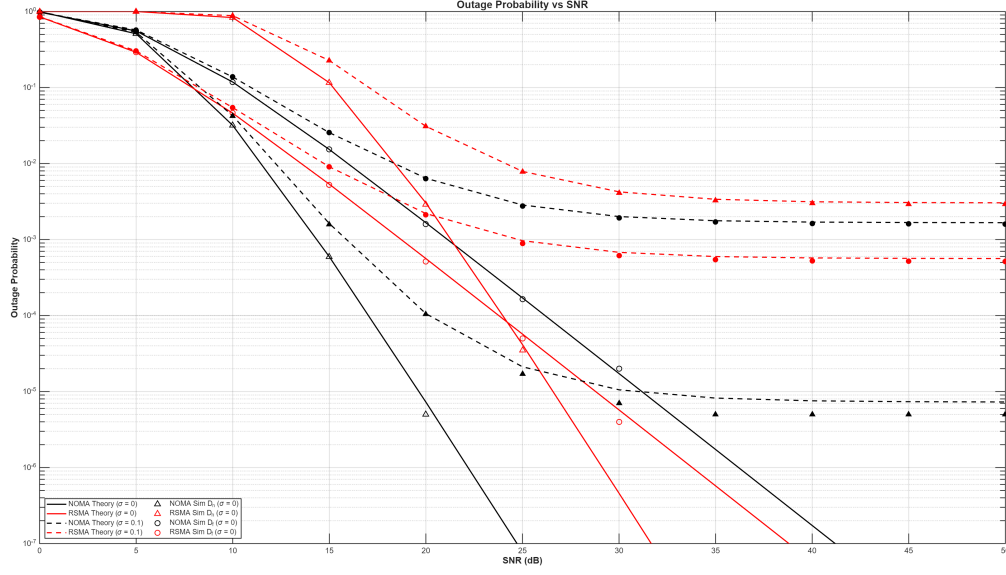


Fig. 1. OP vs SNR with no. of users $K = 2$, no. of antennas $M = N = 2$, ipSIC parameter $\epsilon = 0.01$, and CEEs parameter $\sigma_i = \{0, 0.1\}$

Fig. 2 illustrates the ER vs SNR for both NOMA and RSMA schemes. Here we observe that NOMA performs better than RSMA offering higher ergodic rates.

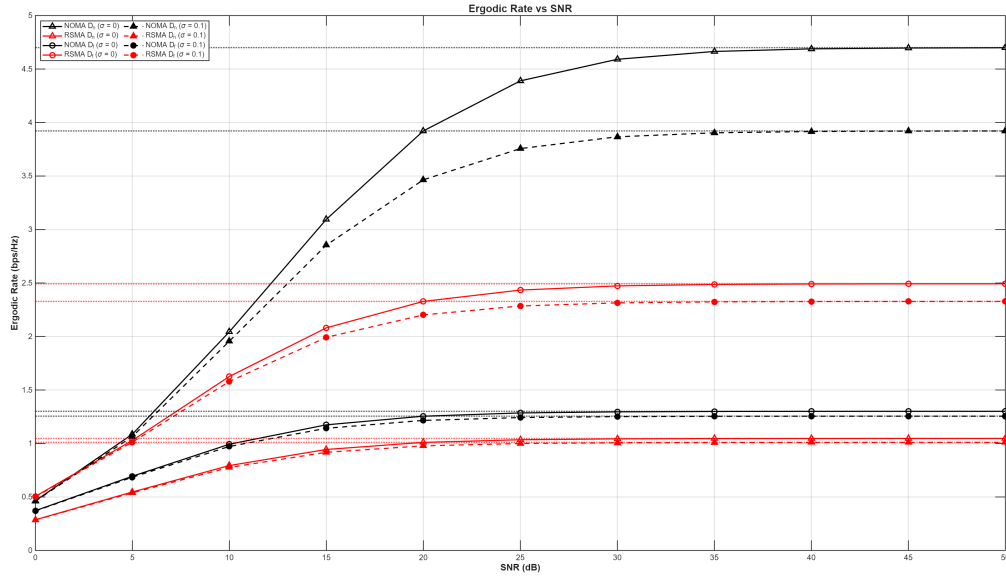


Fig. 2. ER vs SNR with no. of users $K = 2$, no. of antennas $M = N = 2$, ipSIC parameter $\epsilon = 0.01$, and CEEs parameter $\sigma_i = \{0, 0.1\}$

Fig. 3 illustrates PoD vs SNR for both NOMA and RSMA schemes. Here we observe that RSMA offers better PoD compared to NOMA for a given SNR.

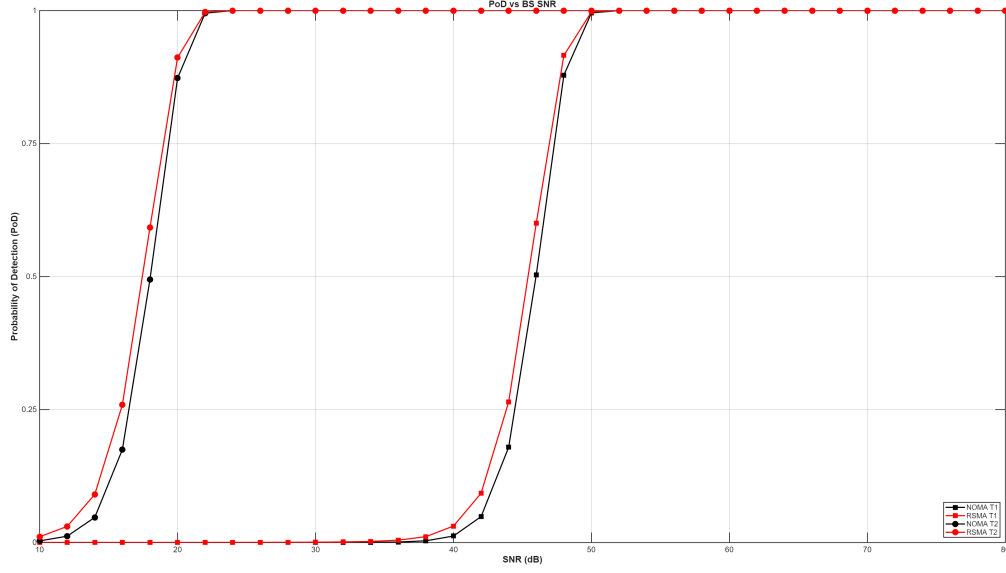


Fig. 3. PoD vs SNR with no. of targets $T = 2$, no. of antennas $M = N = 16$, and PoFA $P_{fa} = 10^{-6}$

Fig. 4 illustrates PoD vs PoFA using Receiver Operating Characteristic (ROC) curves. Since T2 is closer than T1, it shows better detection performance.

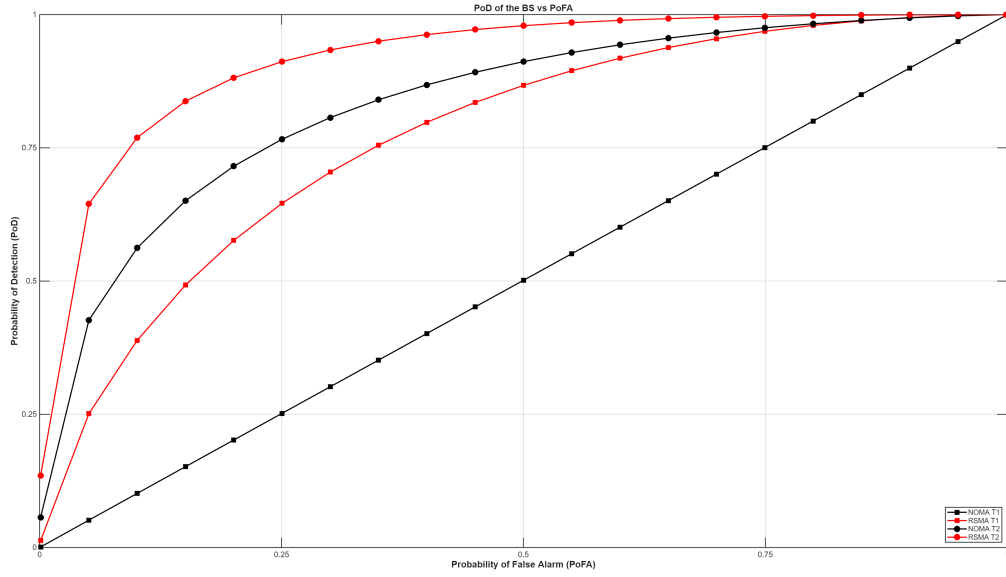


Fig. 4. PoD vs PoFA with no. of targets $T = 2$, no. of antennas $M = N = 16$, and transmit power $P_S = 40$ dBm

Performance Analysis of FD/HD Cooperative NOMA and Proposed RSMA Extensions

A. System model

We consider a three-node cooperative downlink NOMA architecture consisting of a Base Station (BS), a Near User (D_N), and a Far User (D_F). To minimize hardware complexity, all nodes are equipped with a single antenna. The BS communicates directly with D_N , which utilizes a decode-and-forward (DF) protocol to relay data to D_F .

Let h_{ij} denote the fading coefficient between nodes $i \in \{S, N\}$ and $j \in \{N, F\}$. These links are subjected to Nakagami- m fading with a shape parameter m_{ij} and mean power $\mathbb{E}[|h_{ij}|^2] = \sigma_{ij}$. The channel power gains, $|h_{ij}|^2$, follow a Gamma distribution. The probability density function (PDF) is given by:

$$f_{|h_{ij}|^2}(x) = \frac{m_{ij}^{m_{ij}} x^{m_{ij}-1}}{\Gamma(m_{ij}) \sigma_{ij}^{m_{ij}}} e^{-\frac{m_{ij}x}{\sigma_{ij}}}, \quad x \geq 0 \quad (51)$$

For the FD mode, self-loop interference (LI) exists at D_N , represented by a Rayleigh fading channel coefficient h_{LI} with average power Ω_{LI} . We introduce a switching factor $\bar{\omega}$, where $\bar{\omega} = 1$ denotes FD mode and $\bar{\omega} = 0$ denotes HD mode. The superimposed signal transmitted by the BS incorporates power allocation coefficients a_N and a_F (where $a_F > a_N$ and $a_N + a_F = 1$). The received signal at D_N during the m -th time slot is expressed as:

$$y_{D_N}[m] = h_{SN}(\sqrt{a_N P_s} x_N[m] + \sqrt{a_F P_s} x_F[m]) + h_{LI} \sqrt{\bar{\omega} P_r} x_{LI}[m - \tau] + n_{D_N}[m] \quad (52)$$

where P_s and P_r are the transmit powers at the BS and D_N respectively, $x_{LI}[m - \tau]$ is the delayed loop interference signal, and n_{D_N} is additive white Gaussian noise (AWGN).

B. Analysis

1) *SINR Formulations*: According to the NOMA protocol, D_N executes SIC. Because D_F receives higher power allocation (a_F), D_N first decodes D_F 's message (x_F), treats its own message as interference, and subtracts x_F from the total signal to decode x_N .

To simplify the analysis, we adopt the standard cooperative network assumption that the Base Station and the relay node operate with the same average transmit power, such that $P_s = P_r = P$. Let $\rho = P/N_0$ denote the transmit SNR. Consequently, the received Signal-to-Interference-Plus-Noise Ratio (SINR) at D_N to detect x_F is formulated as:

$$\gamma_{D_F \rightarrow D_N} = \frac{|h_{SN}|^2 a_F \rho}{|h_{SN}|^2 a_N \rho + \bar{\omega} |h_{LI}|^2 \rho + 1} \quad (53)$$

Assuming perfect SIC, D_N then decodes its own message x_N with the following SINR:

$$\gamma_{D_N} = \frac{|h_{SN}|^2 a_N \rho}{\bar{\omega} |h_{LI}|^2 \rho + 1} \quad (54)$$

For the Far User (D_F), we analyze two links. The direct link (DL) from the BS provides an SINR bounded by:

$$\gamma_{1,D_F} = \frac{|h_{SF}|^2 a_F \rho}{|h_{SF}|^2 a_N \rho + 1} \quad (55)$$

The relaying link from D_N to D_F yields an SINR bounded by:

$$\gamma_{2,D_F} = |h_{NF}|^2 \rho \quad (56)$$

If D_F combines both the direct and relay signals using Maximum Ratio Combining (MRC), the total received SINR becomes $\gamma_{D_F}^{MRC} = \gamma_{1,D_F} + \gamma_{2,D_F}$. In the "No Direct Link" (NDL) scenario, D_F relies exclusively on γ_{2,D_F} .

2) *Outage Probability Definitions:* An outage occurs if a user cannot decode the desired message at the target rate. Let R_N and R_F be the target rates for D_N and D_F (in BPCU). The required SNR thresholds for FD mode are defined as:

$$\gamma_{th_N}^{FD} = 2^{R_N} - 1, \quad \gamma_{th_F}^{FD} = 2^{R_F} - 1 \quad (57)$$

In HD mode, the transmission requires two time slots, doubling the required spectral efficiency. Thus, the thresholds are:

$$\gamma_{th_N}^{HD} = 2^{2R_N} - 1, \quad \gamma_{th_F}^{HD} = 2^{2R_F} - 1 \quad (58)$$

For D_N , an outage is declared if it fails to detect either x_F or its own message x_N . The probability is denoted by:

$$P_{D_N}^{FD} = 1 - P_r(\gamma_{D_F \rightarrow D_N} > \gamma_{th_F}^{FD}, \gamma_{D_N} > \gamma_{th_N}^{FD}) \quad (59)$$

For D_F in the NDL scenario, an outage occurs if D_N fails to act as a relay (meaning D_N could not decode x_F), OR if D_N successfully relays the signal but D_F fails to decode the relayed transmission:

$$P_{D_F, nodir}^{FD} = P_r(\gamma_{D_F \rightarrow D_N} < \gamma_{th_F}^{FD}) + P_r(\gamma_{2,D_F} < \gamma_{th_F}^{FD}, \gamma_{D_F \rightarrow D_N} > \gamma_{th_F}^{FD}) \quad (60)$$

3) *Delay-Limited System Throughput:* In a delay-limited transmission mode, the system throughput is evaluated based on the source transmission rates and the corresponding probability that the transmission is successfully decoded by the destination (i.e., avoiding an outage). The total system

throughput for FD and HD NOMA configurations is calculated as the sum of the successfully transmitted bit rates for both the Near User and the Far User:

$$R_{sys}^{FD} = (1 - P_{D_N}^{FD})R_N + (1 - P_{D_F}^{FD})R_F \quad (61)$$

$$R_{sys}^{HD} = (1 - P_{D_N}^{HD})R_N + (1 - P_{D_F}^{HD})R_F \quad (62)$$

4) *Ergodic Rate Formulations:* The Ergodic Rate represents the expected value of the instantaneous channel capacity. For the Near User (D_N), the achievable rate is bounded by its ability to decode its own message after successfully executing SIC. In FD mode, this is expressed as:

$$R_{D_N}^{FD} = \mathbb{E} [\log_2(1 + \gamma_{D_N})] \quad (63)$$

For the Far User (D_F), the message x_F must be successfully decoded twice in a DF relay system: first by D_N for relaying, and then by D_F upon reception. Therefore, the ergodic rate of the Far User in the NDL scenario is limited by the weaker of these two links:

$$R_{D_F}^{FD} = \mathbb{E} [\log_2(1 + \min(\gamma_{D_F \rightarrow D_N}, \gamma_{2, D_F}))] \quad (64)$$

In HD mode, the communication process spans two distinct time slots (one for BS-to- D_N transmission, and one for D_N -to- D_F relaying). Because the same amount of information takes twice the time to transmit, the achievable ergodic rates for both users are scaled by a factor of one-half:

$$R_{D_N}^{HD} = \frac{1}{2} \mathbb{E} [\log_2(1 + \gamma_{D_N})] \quad (65)$$

$$R_{D_F}^{HD} = \frac{1}{2} \mathbb{E} [\log_2(1 + \min(\gamma_{D_F \rightarrow D_N}, \gamma_{2, D_F}))] \quad (66)$$

C. Results and Discussions

A Monte Carlo simulation consisting of 10^6 iterations was developed in MATLAB based on the system model. The results are analyzed across three primary performance metrics: Outage Probability, Delay-Limited System Throughput, and Ergodic Rate.

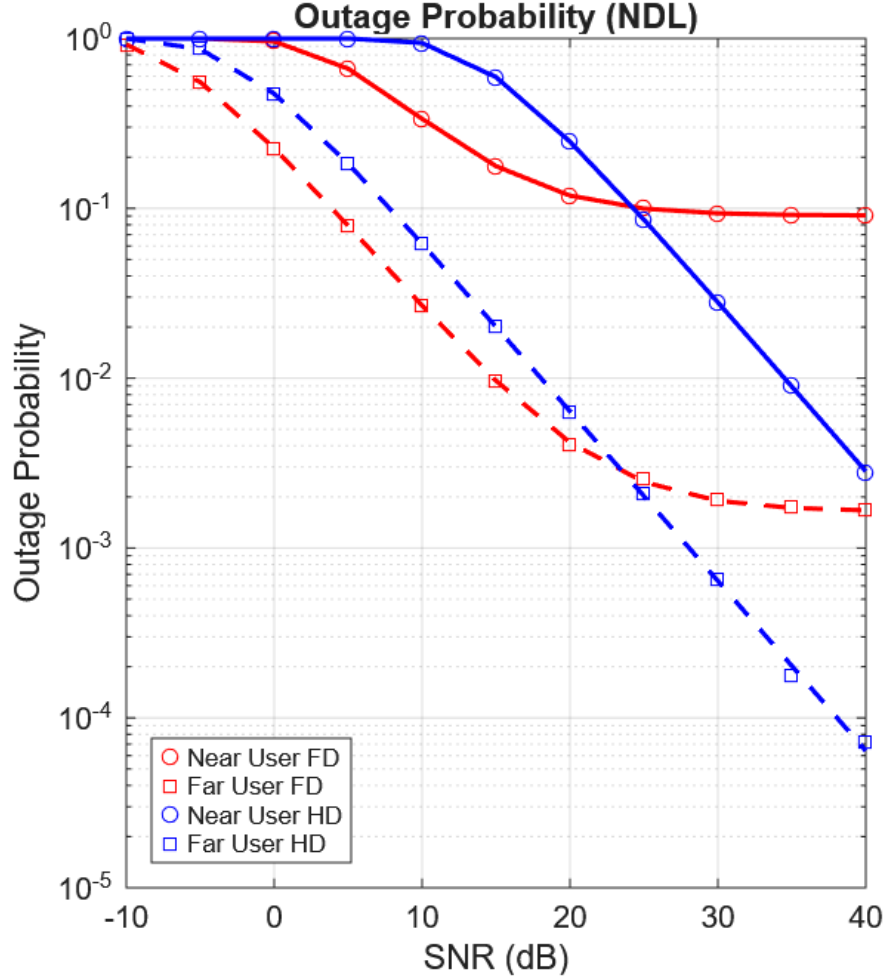


Fig. 5. Outage Probability versus Transmit SNR for FD and HD NOMA without a Direct Link.

As depicted in Fig. 5, the outage probability curves illustrate the stark contrast between the two transmission modes. At lower SNRs (-10 dB to approximately 15 dB), FD-NOMA (red curves) demonstrates lower outage probabilities, offering superior reliability compared to HD-NOMA (blue curves). However, as transmit power (ρ) increases beyond 20 dB, both FD-NOMA outage curves flatline, converging to horizontal error floors (around 10^{-1} for the Near User and 2×10^{-3} for the Far User). This confirms the "zero-diversity order" caused by self-loop interference permanently bottlenecking the SINR. In contrast, HD-NOMA maintains a strict downward trajectory, with the Far User HD outage falling below 10^{-4} , showcasing significantly better reliability at high SNRs.

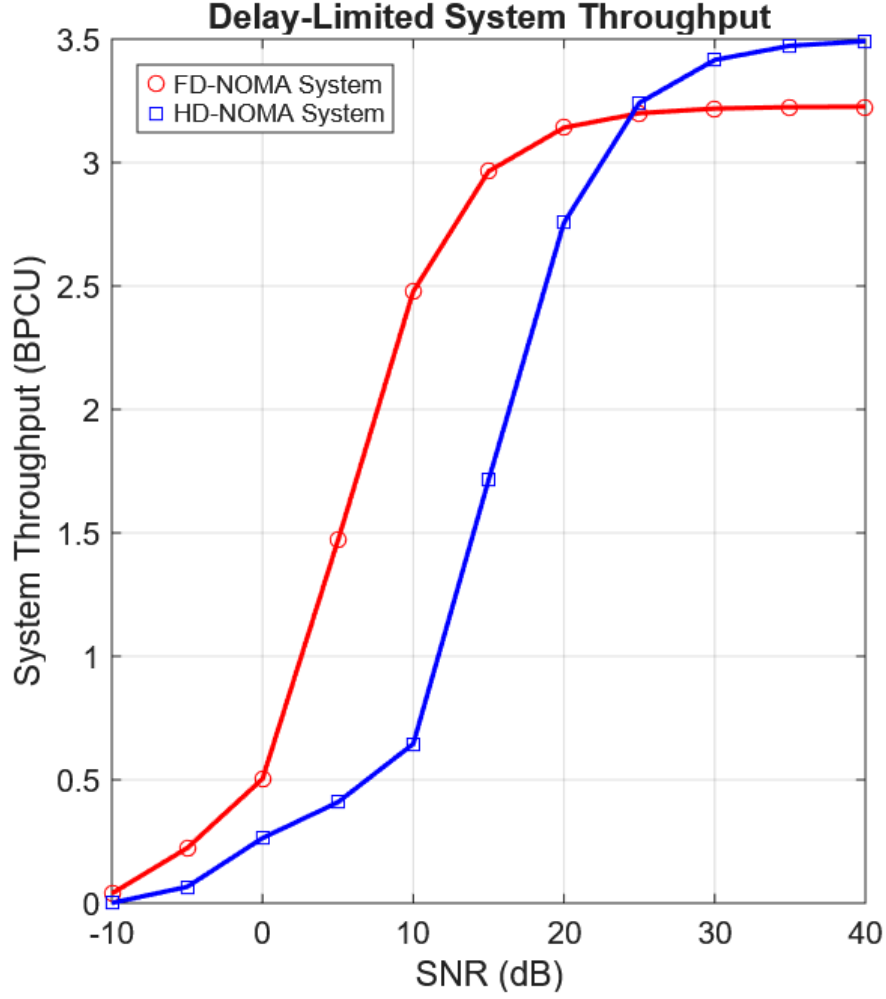


Fig. 6. Delay-Limited System Throughput versus Transmit SNR.

Fig. 6 illustrates the Delay-Limited System Throughput. The graph visualizes a distinct operational crossover point just before 30 dB (approximately 25 dB). Below this threshold, FD-NOMA (red circles) exhibits a massive throughput advantage due to completing the cooperative transmission in a single time slot. However, beyond 15 dB, FD-NOMA's throughput growth stalls, hitting a hard ceiling at approximately 3.2 BPCU due to its outage error floor. The 2-slot HD-NOMA system (blue squares), unhindered by loop interference, surges past FD-NOMA and eventually peaks at 3.5 BPCU, demonstrating that FD-NOMA is optimal strictly for low-to-medium power environments.

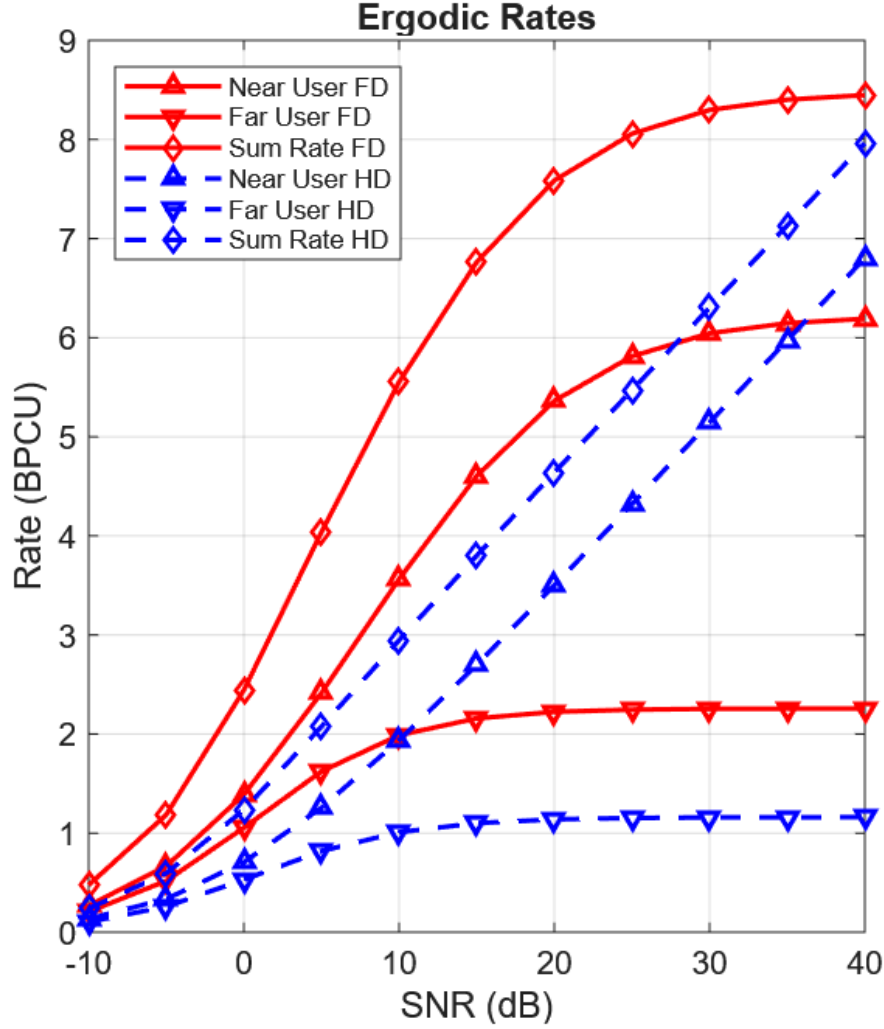


Fig. 7. Ergodic Sum Rates versus Transmit SNR for FD and HD NOMA.

Finally, Fig. 7 displays the Ergodic Rates for the individual users and the total system sum rate. The impact of interference causes severe capacity ceilings for the Far Users in both modes. The Far User FD (solid red, downward triangles) hits a ceiling at approximately 2.2 BPCU, while the Far User HD (dashed blue, downward triangles) plateaus even lower, near 1.1 BPCU. Despite this, the FD Sum Rate (red diamonds) initially dominates due to its single-slot transmission efficiency. However, at higher SNRs, the Near User HD rate (dashed blue, upward triangles) grows aggressively, pulling the HD Sum Rate up linearly. By 40 dB, the HD Sum Rate is rapidly closing the gap with the FD Sum Rate, reinforcing the need for advanced interference mitigation techniques to unlock higher capacities for all users in densely deployed networks.

III. CONCLUSIONS

In this paper, we presented a comprehensive comparative analysis of NOMA and RSMA frameworks under Nakagami-m fading, evaluating their performance across integrated sensing and communication (ISAC) networks and cooperative relaying systems. Through our analysis of the downlink ISAC system with imperfect SIC, we observed a distinct performance trade-off between the two multiple-access schemes. Simulations demonstrated that RSMA offers superior outage probability performance for the far user and achieves a higher Probability of Detection (PoD) for sensing targets at a given SNR. Conversely, NOMA proved more resilient for the near user's outage probability and consistently delivered higher ergodic rates across the system.

Furthermore, our investigation into cooperative decode-and-forward architectures highlighted the fundamental limitations of Full-Duplex (FD) operations. While FD-NOMA provides a massive throughput advantage at low-to-medium SNRs due to single-slot transmission, it inevitably hits an error floor at higher transmit powers due to self-loop interference. Half-Duplex (HD) NOMA, unhindered by this interference, scales much more effectively, ultimately achieving higher peak throughput and superior reliability in high-power regimes.

Finally, it is important to note that during the reproduction and validation of the theoretical models, we identified a few mathematical discrepancies within the original reference literature. Specifically, an operational error was found in the referenced formulas where a maximum (max) function was utilized instead of the correct minimum (min) function for determining specific bounds. Additionally, explicit modeling for the signal attenuation was missing from the foundational text; therefore, we integrated our own assumed attenuation model to successfully execute the Monte Carlo simulations and complete the comparative analysis. Despite these adjustments, the overarching performance trends remain consistent, underscoring the critical need to balance capacity limits, user fairness, and interference management in future network deployments.

IV. APPENDIX

NOMA vs RSMA in Downlink ISAC with Imperfect SIC

1) *Communication:* The following MATLAB code was developed to compare the OPs and ERs performance of NOMA and RSMA.

```
clear; clc; close all;
```

```

%% =====
% 1. SYSTEM PARAMETERS
% =====
% Simulation & SNR
N          = 1e6;           % Monte-Carlo iterations
N0         = 1;             % Noise variance
P_S_dB     = 0:5:50;        % Transmit power in dB
P_S        = 10.^((P_S_dB)./10); % Transmit power in linear scale
gamma_S    = P_S ./ N0;     % Transmit SNR

% Channel Parameters (Nakagami-m)
m_SDn      = 2;             % Shape factor (Near User)
m_SDf      = 1;             % Shape factor (Far User)
Omega_SDn   = 1;            % Spread factor (Near User)
Omega_SDf   = 1;            % Spread factor (Far User)

% Target SINR Thresholds
gamma_thr   = 0.5;
gamma_thn   = 0.6;
gamma_thf   = 0.8;
gamma_thc   = 0.6;
gamma_thnp  = 0.26;
gamma_thfp  = 0.42;

% Hardware Impairment & Power Allocation
epsilon     = 0.01;         % Hardware impairment level
sigma_SDf   = [0; 0.1];     % Far user hardware noise [Ideal; Impaired]
sigma_SDn   = [0; 0.1];     % Near user hardware noise [Ideal; Impaired]

% Power Coefficients (RSMA & NOMA)
a_c         = 0.3;          % RSMA Common
a_np        = 0.05;         % RSMA Near Private
a_fp        = 0.15;         % RSMA Far Private
a_r         = 0.5;          % Radar/Sensing
a_n         = 0.2;          % NOMA Near
a_f         = 0.3;          % NOMA Far

M = 2;           % Number of Antennas

%% =====
% 2. CHANNEL GENERATION
% =====
g_SDf = gamrnd(m_SDf*M, Omega_SDf/m_SDf, 1, N);
g_SDn = gamrnd(m_SDn*M, Omega_SDn/m_SDn, 1, N);

%% =====
% 3. THEORETICAL OUTAGE PROBABILITY (OP)
% =====

```

```

% --- RSMA Denominators (Interference Limits) ---
den_RS_r = a_r - (a_np + a_fp + a_c)*gamma_thr;
den_RS_c = a_c - (a_np + a_fp + epsilon*a_r)*gamma_thc;
den_RS_fp = a_fp - (a_np + epsilon*(a_c + a_r))*gamma_thfp;
den_RS_np = a_np - (a_fp + epsilon*(a_c + a_r))*gamma_thnp;

% --- RSMA Thresholds (phi -> Far, psi -> Near) ---
phi_r = gamma_thr * (sigma_SDf.^2 + 1./gamma_S) / den_RS_r;
phi_c = gamma_thc * (sigma_SDf.^2 + 1./gamma_S) / den_RS_c;
phi_fp = gamma_thfp * (sigma_SDf.^2 + 1./gamma_S) / den_RS_fp;

psi_r = gamma_thr * (sigma_SDn.^2 + 1./gamma_S) / den_RS_r;
psi_c = gamma_thc * (sigma_SDn.^2 + 1./gamma_S) / den_RS_c;
psi_np = gamma_thnp * (sigma_SDn.^2 + 1./gamma_S) / den_RS_np;

% Impossible SINR targets (Negative Denominators -> Outage = 1)
phi_r(phi_r < 0) = Inf; phi_c(phi_c < 0) = Inf; phi_fp(phi_fp < 0) = Inf;
psi_r(psi_r < 0) = Inf; psi_c(psi_c < 0) = Inf; psi_np(psi_np < 0) = Inf;

% Max thresholds for OP calculation
phi = max(max(phi_r, phi_c), phi_fp);
psi = max(max(psi_r, psi_c), psi_np);

P_RS_Df = gammainc((m_SDf*phi)/Omega_SDf, m_SDf*M, 'lower');
P_RS_Dn = gammainc((m_SDn*psi)/Omega_SDn, m_SDn*M, 'lower');

% --- NOMA Denominators ---
den_NO_1 = a_r - (a_n + a_f)*gamma_thr;
den_NO_2 = a_f - (a_n + epsilon*a_r)*gamma_thf;
den_NO_3 = a_n - epsilon*(a_f + a_r)*gamma_thn;

% --- NOMA Thresholds ---
phi_d_1 = gamma_thr * (sigma_SDf.^2 + 1./gamma_S) / den_NO_1;
phi_d_2 = gamma_thf * (sigma_SDf.^2 + 1./gamma_S) / den_NO_2;

psi_d_1 = gamma_thr * (sigma_SDn.^2 + 1./gamma_S) / den_NO_1;
psi_d_2 = gamma_thf * (sigma_SDn.^2 + 1./gamma_S) / den_NO_2;
psi_d_3 = gamma_thn * (sigma_SDn.^2 + 1./gamma_S) / den_NO_3;

phi_d_1(phi_d_1 < 0) = Inf; phi_d_2(phi_d_2 < 0) = Inf;
psi_d_1(psi_d_1 < 0) = Inf; psi_d_2(psi_d_2 < 0) = Inf; psi_d_3(psi_d_3 < 0) = Inf;

% Max thresholds
phi_d = max(phi_d_1, phi_d_2);
psi_d = max(max(psi_d_1, psi_d_2), psi_d_3);

P_NO_Df = gammainc((m_SDf*phi_d)/Omega_SDf, m_SDf*M, 'lower');
P_NO_Dn = gammainc((m_SDn*psi_d)/Omega_SDn, m_SDn*M, 'lower');

```

```

%% =====
% 4. THEORETICAL ASYMPTOTIC ER (High SNR Ceiling)
% =====
% --- NOMA Asymptotic Limits ---
SINR_NO_xn_inf = a_n ./ (epsilon*(a_r + a_f) + sigma_SDn.^2 ./ g_SDn);
R_NO_Dn_h = mean(log2(1 + SINR_NO_xn_inf), 2);

SINR_NO_xf_SDn_inf = a_f ./ (a_n + epsilon*a_r + sigma_SDn.^2 ./ g_SDn);
SINR_NO_xf_SDF_inf = a_f ./ (a_n + epsilon*a_r + sigma_SDF.^2 ./ g_SDF);
R_NO_Df_h = mean(log2(1 + min(SINR_NO_xf_SDn_inf, SINR_NO_xf_SDF_inf)), 2);

% --- RSMA Asymptotic Limits ---
SINR_RS_xc_SDn_inf = a_c ./ (a_np + a_fp + epsilon*a_r + sigma_SDn.^2 ./ g_SDn);
SINR_RS_xc_SDF_inf = a_c ./ (a_np + a_fp + epsilon*a_r + sigma_SDF.^2 ./ g_SDF);
R_RS_C_h = mean(log2(1 + min(SINR_RS_xc_SDn_inf, SINR_RS_xc_SDF_inf)), 2);

SINR_RS_xnp_inf = a_np ./ (epsilon*(a_c + a_r) + a_fp + sigma_SDn.^2 ./ g_SDn);
R_RS_Dn_h = (R_RS_C_h / 2) + mean(log2(1 + SINR_RS_xnp_inf), 2);

SINR_RS_xfp_inf = a_fp ./ (epsilon*(a_c + a_r) + a_np + sigma_SDF.^2 ./ g_SDF);
R_RS_Df_h = (R_RS_C_h / 2) + mean(log2(1 + SINR_RS_xfp_inf), 2);

%% =====
% 5. MONTE-CARLO SIMULATIONS
% =====
num_snr = numel(P_S);
P_out_NO_Dn = zeros(2, num_snr);
P_out_NO_Df = zeros(2, num_snr);
P_out_RS_Dn = zeros(2, num_snr);
P_out_RS_Df = zeros(2, num_snr);

R_NO_Dn = zeros(2, num_snr);
R_NO_Df = zeros(2, num_snr);
R_RS_Dn = zeros(2, num_snr);
R_RS_Df = zeros(2, num_snr);

disp('Running_Monte-Carlo_Loop...');
for i = 1:num_snr
    g_S = gamma_S(i);

    % -----
    % NOMA Simulation
    % -----
    % Near User
    gamma_NO_xr_SDn = (g_SDn*a_r*g_S) ./ ...
        ((g_SDn*a_n + g_SDn*a_f + sigma_SDn.^2)*g_S + 1);
    gamma_NO_xf_SDn = (g_SDn*a_f*g_S) ./ ...

```

```

        ((g_SDn*a_n + g_SDn*epsilon*a_r + sigma_SDn.^2)*g_S + 1);
gamma_NO_xn_SDn = (g_SDn*a_n*g_S) ./ ...
        ((g_SDn*a_f + g_SDn*a_r)*epsilon*g_S + sigma_SDn.^2*g_S + 1);

NO_Dn_success = (gamma_NO_xr_SDn > gamma_thr) & ...
        (gamma_NO_xf_SDn > gamma_thf) & ...
        (gamma_NO_xn_SDn > gamma_thn);
P_out_NO_Dn(:, i) = 1 - mean(NO_Dn_success, 2);
R_NO_Dn(:, i)      = mean(log2(1 + gamma_NO_xn_SDn), 2);

% Far User
gamma_NO_xr_SDf = (g_SDf*a_r*g_S) ./ ...
        ((g_SDf*a_n + g_SDf*a_f + sigma_SDf.^2)*g_S + 1);
gamma_NO_xf_SDf = (g_SDf*a_f*g_S) ./ ...
        ((g_SDf*a_n + g_SDf*epsilon*a_r + sigma_SDf.^2)*g_S + 1);

NO_Df_success = (gamma_NO_xr_SDf > gamma_thr) & ...
        (gamma_NO_xf_SDf > gamma_thf);
P_out_NO_Df(:, i) = 1 - mean(NO_Df_success, 2);
R_NO_Df(:, i)      = mean(log2(1 + min(gamma_NO_xf_SDn, gamma_NO_xf_SDf)), 2);

% -----
% RSMA Simulation
% -----

% Near User
gamma_RS_xr_SDn = (a_r*g_SDn*g_S) ./ ...
        ((a_np*g_SDn + a_fp*g_SDn + a_c*g_SDn + sigma_SDn.^2)*g_S + 1);
gamma_RS_xc_SDn = (a_c*g_SDn*g_S) ./ ...
        ((a_np*g_SDn + a_fp*g_SDn + epsilon*a_r*g_SDn + sigma_SDn.^2)*g_S + 1);
gamma_RS_xnp_SDn = (a_np*g_SDn*g_S) ./ ...
        ((epsilon*a_c*g_SDn + a_fp*g_SDn + epsilon*a_r*g_SDn + sigma_SDn.^2)*g_S + 1);

RS_Dn_success = (gamma_RS_xr_SDn > gamma_thr) & ...
        (gamma_RS_xc_SDn > gamma_thc) & ...
        (gamma_RS_xnp_SDn > gamma_thnp);
P_out_RS_Dn(:, i) = 1 - mean(RS_Dn_success, 2);

% Far User
gamma_RS_xr_SDf = (a_r*g_SDf*g_S) ./ ...
        ((a_np*g_SDf + a_fp*g_SDf + a_c*g_SDf + sigma_SDf.^2)*g_S + 1);
gamma_RS_xc_SDf = (a_c*g_SDf*g_S) ./ ...
        ((a_np*g_SDf + a_fp*g_SDf + epsilon*a_r*g_SDf + sigma_SDf.^2)*g_S + 1);
gamma_RS_xfp_SDf = (a_fp*g_SDf*g_S) ./ ...
        ((epsilon*a_c*g_SDf + a_np*g_SDf + epsilon*a_r*g_SDf + sigma_SDf.^2)*g_S + 1);

RS_Df_success = (gamma_RS_xr_SDf > gamma_thr) & ...
        (gamma_RS_xc_SDf > gamma_thc) & ...
        (gamma_RS_xfp_SDf > gamma_thfp);

```

```

P_out_RS_Df(:, i) = 1 - mean(RS_Df_success, 2);

% RSMA Ergodic Rates
RS_C_ER = min(log2(1 + gamma_RS_xc_SDn), log2(1 + gamma_RS_xc_SDf));
R_RS_Dn(:, i) = mean((RS_C_ER/2) + log2(1 + gamma_RS_xnp_SDn), 2);
R_RS_Df(:, i) = mean((RS_C_ER/2) + log2(1 + gamma_RS_xfp_SDf), 2);
end
disp('Simulation_Complete.');
```

%% =====

% 6. PLOTTING: OUTAGE PROBABILITY

% =====

```

figure('Name', 'Outage_Probability', 'Position', [100, 100, 800, 600]);

% --- Case 1: Ideal Hardware (sigma = 0) ---
h_NO_th_0 = semilogy(P_S_dB, P_NO_Df(1,:), 'k-', 'LineWidth', 1.8); hold on;
            semilogy(P_S_dB, P_NO_Dn(1,:), 'k-', 'LineWidth', 1.8);
h_NO_dn_0 = semilogy(P_S_dB, P_out_NO_Dn(1,:), 'k^', 'MarkerSize', 8, 'LineWidth', 1.2);
h_NO_df_0 = semilogy(P_S_dB, P_out_NO_Df(1,:), 'ko', 'MarkerSize', 8, 'LineWidth', 1.2);

h_RS_th_0 = semilogy(P_S_dB, P_RS_Df(1,:), 'r-', 'LineWidth', 1.8);
            semilogy(P_S_dB, P_RS_Dn(1,:), 'r-', 'LineWidth', 1.8);
h_RS_dn_0 = semilogy(P_S_dB, P_out_RS_Dn(1,:), 'r^', 'MarkerSize', 8, 'LineWidth', 1.2);
h_RS_df_0 = semilogy(P_S_dB, P_out_RS_Df(1,:), 'ro', 'MarkerSize', 8, 'LineWidth', 1.2);

% --- Case 2: Impaired Hardware (sigma = 0.1) ---
h_NO_th_1 = semilogy(P_S_dB, P_NO_Df(2,:), 'k--', 'LineWidth', 1.8);
            semilogy(P_S_dB, P_NO_Dn(2,:), 'k--', 'LineWidth', 1.8);
h_NO_dn_1 = semilogy(P_S_dB, P_out_NO_Dn(2,:), 'k^', 'MarkerFaceColor', 'k', 'MarkerSize', 8);
h_NO_df_1 = semilogy(P_S_dB, P_out_NO_Df(2,:), 'ko', 'MarkerFaceColor', 'k', 'MarkerSize', 8);

h_RS_th_1 = semilogy(P_S_dB, P_RS_Df(2,:), 'r--', 'LineWidth', 1.8);
            semilogy(P_S_dB, P_RS_Dn(2,:), 'r--', 'LineWidth', 1.8);
h_RS_dn_1 = semilogy(P_S_dB, P_out_RS_Dn(2,:), 'r^', 'MarkerFaceColor', 'r', 'MarkerSize', 8);
h_RS_df_1 = semilogy(P_S_dB, P_out_RS_Df(2,:), 'ro', 'MarkerFaceColor', 'r', 'MarkerSize', 8);

% Formatting
ylim([1e-7, 1]); xlim([0, 50]);
grid on; set(gca, 'FontSize', 12, 'LineWidth', 1.2);
xlabel('SNR_(dB)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Outage_Probability', 'FontSize', 14, 'FontWeight', 'bold');
title('Outage_Probability_vs_SNR', 'FontSize', 16);

% OP Legend
lqd_op = legend([h_NO_th_0, h_RS_th_0, h_NO_th_1, h_RS_th_1, ...
                h_NO_dn_0, h_RS_dn_0, h_NO_df_0, h_RS_df_0], ...
                {'NOMA_Theory_(\sigma=_0)', 'RSMA_Theory_(\sigma=_0)', ...
                'NOMA_Theory_(\sigma=_0.1)', 'RSMA_Theory_(\sigma=_0.1)', ...
```



```

        'NOMA_Sim_D_n_(\sigma=_0)', 'RSMA_Sim_D_n_(\sigma=_0)', ...
        'NOMA_Sim_D_f_(\sigma=_0)', 'RSMA_Sim_D_f_(\sigma=_0)'}}, ...
    'Location', 'southwest', 'NumColumns', 2, 'FontSize', 11);

%% =====
% 7. PLOTTING: ERGODIC RATE
% =====

figure('Name', 'Ergodic_Rate', 'Position', [920, 100, 800, 600]);

% --- Case 1: Ideal Hardware (sigma = 0) ---
h_er_NO_dn_0 = plot(P_S_dB, R_NO_Dn(1,:), 'k-^', 'LineWidth', 1.8, 'MarkerSize', 8); hold on;
h_er_NO_df_0 = plot(P_S_dB, R_NO_Df(1,:), 'k-o', 'LineWidth', 1.8, 'MarkerSize', 8);
h_er_RS_dn_0 = plot(P_S_dB, R_RS_Dn(1,:), 'r-^', 'LineWidth', 1.8, 'MarkerSize', 8);
h_er_RS_df_0 = plot(P_S_dB, R_RS_Df(1,:), 'r-o', 'LineWidth', 1.8, 'MarkerSize', 8);

% --- Case 2: Impaired Hardware (sigma = 0.1) ---
h_er_NO_dn_1 = plot(P_S_dB, R_NO_Dn(2,:), 'k--^', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k');
h_er_NO_df_1 = plot(P_S_dB, R_NO_Df(2,:), 'k--o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k');
h_er_RS_dn_1 = plot(P_S_dB, R_RS_Dn(2,:), 'r--^', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');
h_er_RS_df_1 = plot(P_S_dB, R_RS_Df(2,:), 'r--o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');

% --- Asymptotic Lines ---
yline(R_NO_Dn_h(1), 'k:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_RS_Dn_h(1), 'r:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_NO_Dn_h(2), 'k:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_RS_Dn_h(2), 'r:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');

yline(R_NO_Df_h(1), 'k:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_RS_Df_h(1), 'r:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_NO_Df_h(2), 'k:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');
yline(R_RS_Df_h(2), 'r:', 'LineWidth', 2, 'Alpha', 0.6, 'HandleVisibility', 'off');

% Formatting
xlim([0, 50]);
grid on; set(gca, 'FontSize', 12, 'LineWidth', 1.2);
xlabel('SNR_(dB)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Ergodic_Rate_(bps/Hz)', 'FontSize', 14, 'FontWeight', 'bold');
title('Ergodic_Rate_vs_SNR', 'FontSize', 16);

% ER Legend
lgd_er = legend([h_er_NO_dn_0, h_er_RS_dn_0, h_er_NO_df_0, h_er_RS_df_0, ...
                h_er_NO_dn_1, h_er_RS_dn_1, h_er_NO_df_1, h_er_RS_df_1], ...
    {'NOMA_D_n_(\sigma=_0)', 'RSMA_D_n_(\sigma=_0)', ...
    'NOMA_D_f_(\sigma=_0)', 'RSMA_D_f_(\sigma=_0)', ...
    'NOMA_D_n_(\sigma=_0.1)', 'RSMA_D_n_(\sigma=_0.1)', ...
    'NOMA_D_f_(\sigma=_0.1)', 'RSMA_D_f_(\sigma=_0.1)'}}, ...
    'Location', 'northwest', 'NumColumns', 2, 'FontSize', 11);

```

2) *Sensing*: The following MATLAB code was developed to compare the PoD performance of NOMA and RSMA.

```
clear; clc; close all;

%% =====
% 1. SHARED SENSING PARAMETERS & TARGET SETUP
% =====

N0 = 1;

% Antennas & Arrays
M = 16; % Transmit antennas
N = 16; % Receive antennas
lambda = 0.01;
d = lambda / 2;

% Target Setup
theta_deg = [10, 45];
theta = theta_deg * pi / 180;
dist = [50, 10]; % True physical distances in meters
K = 2;

% Power allocations
a_r = 0.5; a_c = 0.3; a_np = 0.05; a_fp = 0.15; % RSMA
a_n = 0.2; a_f = 0.3; % NOMA

% -----
% PHYSICALLY CORRECT RADAR PATH LOSS
% -----
% Radar echo power decays proportionally to 1 / d^4
scale_factor = 1e4;
beta_squared = scale_factor ./ (dist.^4 * M * N);
beta_k = sqrt(beta_squared);

% Build Steering Vectors & Target Response Matrix (G)
G_k = zeros(N, M, K);
R_t_all = cell(K, 1);

for t = 1:K
    a_rx = exp(1j * 2*pi*d/lambda * (0:N-1)' * sin(theta(t)));
    b_tx = exp(1j * 2*pi*d/lambda * (0:M-1)' * sin(theta(t)));
    G_k(:, :, t) = beta_k(t) * a_rx * b_tx.';

    % Store the isolated spatial covariance matrix for each target
    R_t_all{t} = G_k(:, :, t) * G_k(:, :, t)';
end

% Optimizer options for root finding
options = optimset('Display', 'off', 'TolX', 1e-4);
```

```

%% =====
% 2. CALCULATE Figure 8 (PoD vs SNR)
% =====
disp('Calculating_PoD_vs_SNR...');

% Expanded SNR range to capture both the strong and weak targets!
P_S_dB = 10:2:80;
P_S = 10.^(P_S_dB./10);
P_fa_target = 1e-5;

PoD_RS_fig8 = zeros(K, length(P_S));
PoD_NO_fig8 = zeros(K, length(P_S));

for t = 1:K
    R_t = R_t_all{t};
    for i = 1:length(P_S)
        curr_PS = P_S(i);

        % --- RSMA ---
        ncp_fa_RS = sqrt(abs(trace((a_np + a_fp)*curr_PS * R_t)) / N0);
        obj_fa_RS = @(xi) marcumq(ncp_fa_RS, sqrt(2*xi/N0), 3) - P_fa_target;
        xi_RS = fzero(obj_fa_RS, [0, 1e8], options);
        ncp_d_RS = sqrt(abs(trace(curr_PS * R_t)) / N0);
        PoD_RS_fig8(t, i) = marcumq(ncp_d_RS, sqrt(2*xi_RS/N0), 5);

        % --- NOMA ---
        ncp_fa_NO = sqrt(abs(trace(a_n * curr_PS * R_t)) / N0);
        obj_fa_NO = @(xi) marcumq(ncp_fa_NO, sqrt(2*xi/N0), 3) - P_fa_target;
        xi_NO = fzero(obj_fa_NO, [0, 1e8], options);
        ncp_d_NO = sqrt(abs(trace(curr_PS * R_t)) / N0);
        PoD_NO_fig8(t, i) = marcumq(ncp_d_NO, sqrt(2*xi_NO/N0), 3);
    end
end

%% =====
% 3. CALCULATE Figure 9 (PoD vs PoFA - ROC Curves)
% =====
disp('Calculating_PoD_vs_PoFA_at_40_dBm...');

% 1. CORRECT POWER CONVERSION: 40 dBm -> Watts
P_S_dBm = 40; % Power in dBm
P_S_W = 10^((P_S_dBm - 30)/10); % Convert dBm to Watts (10 Watts)
curr_PS = P_S_W; % Linear transmit power (assuming N0 = 1W)

PoFA_vec = linspace(0.001, 0.999, 21);

PoD_RS_fig9 = zeros(K, length(PoFA_vec));

```

```

PoD_NO_fig9 = zeros(K, length(PoFA_vec));

for t = 1:K
    R_t = R_t_all{t};

    % --- RSMA Non-centrality parameters ---
    ncp_fa_RS = sqrt(abs(trace((a_np + a_fp)*curr_PS * R_t)) / N0);
    ncp_d_RS = sqrt(abs(trace(curr_PS * R_t)) / N0);

    % --- NOMA Non-centrality parameters ---
    ncp_fa_NO = sqrt(abs(trace(a_n * curr_PS * R_t)) / N0);
    ncp_d_NO = sqrt(abs(trace(curr_PS * R_t)) / N0);

    for i = 1:length(PoFA_vec)
        curr_PoFA = PoFA_vec(i);

        % --- RSMA Detection ---
        obj_fa_RS = @(xi) marcumq(ncp_fa_RS, sqrt(2*xi/N0), 3) - curr_PoFA;
        xi_RS = fzero(obj_fa_RS, [0, 1e8], options);
        PoD_RS_fig9(t, i) = marcumq(ncp_d_RS, sqrt(2*xi_RS/N0), 5);

        % --- NOMA Detection ---
        obj_fa_NO = @(xi) marcumq(ncp_fa_NO, sqrt(2*xi/N0), 3) - curr_PoFA;
        xi_NO = fzero(obj_fa_NO, [0, 1e8], options);
        PoD_NO_fig9(t, i) = marcumq(ncp_d_NO, sqrt(2*xi_NO/N0), 3);
    end
end

disp('Calculations_Complete!');

%% =====
% 4. PLOTTING
% =====

% --- FIGURE 8 ---
figure('Name', 'Physically_Correct:_PoD_vs_SNR', 'Position', [100, 100, 700, 500]);
plot(P_S_dB, PoD_NO_fig8(1,:), 'k-s', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k'); hold on;
plot(P_S_dB, PoD_RS_fig8(1,:), 'r-s', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');
plot(P_S_dB, PoD_NO_fig8(2,:), 'k-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k');
plot(P_S_dB, PoD_RS_fig8(2,:), 'r-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');

xlim([10, 80]); ylim([0, 1]); yticks(0:0.25:1.00);
grid on; set(gca, 'FontSize', 12, 'LineWidth', 1.2);
xlabel('SNR_(dB)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Probability_of_Detection_(PoD)', 'FontSize', 14, 'FontWeight', 'bold');
title('PoD_vs_BS_SNR', 'FontSize', 14);
legend('NOMA_T1', 'RSMA_T1', 'NOMA_T2', 'RSMA_T2', ...
    'Location', 'southeast', 'FontSize', 11);

% --- FIGURE 9 ---

```

```

figure('Name', 'Physically_Correct:_ROC_Curves', 'Position', [820, 100, 700, 500]);
plot(PoFA_vec, PoD_NO_fig9(1,:), 'k-s', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k'); hold on;
plot(PoFA_vec, PoD_RS_fig9(1,:), 'r-s', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');
plot(PoFA_vec, PoD_NO_fig9(2,:), 'k-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'k');
plot(PoFA_vec, PoD_RS_fig9(2,:), 'r-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'MarkerFaceColor', 'r');

xlim([0, 1]); ylim([0, 1]); xticks(0:0.25:1.00); yticks(0:0.25:1.00);
grid on; set(gca, 'FontSize', 12, 'LineWidth', 1.2);
xlabel('Probability_of_False_Alarm_(PoFA)', 'FontSize', 14, 'FontWeight', 'bold');
ylabel('Probability_of_Detection_(PoD)', 'FontSize', 14, 'FontWeight', 'bold');
title('PoD_of_the_BS_vs_PoFA', 'FontSize', 14);
legend('NOMA_T1', 'RSMA_T1', ...
        'NOMA_T2', 'RSMA_T2', ...
        'Location', 'southeast', 'FontSize', 11);

```

Performance Analysis of FD/HD Cooperative NOMA and Proposed RSMA Extensions

```

clear; clc; close all;

%% 1. System Parameters
N_sim = 1e6;
SNR_dB = -10:5:40;
rho_array = 10.^(SNR_dB/10);

alpha = 2;
d = 0.3;
sigma_SN = d^(-alpha);
sigma_NF = (1-d)^(-alpha);

a_N = 0.2;
a_F = 0.8;
m = 1;

omega_LI_dB = -15;
omega_LI = 10^(omega_LI_dB/10);

R_N = 3;
R_F = 0.5;

gamma_th_N_FD = 2^(R_N) - 1;
gamma_th_F_FD = 2^(R_F) - 1;

gamma_th_N_HD = 2^(2*R_N) - 1;
gamma_th_F_HD = 2^(2*R_F) - 1;

%% 2. Initialize Storage Arrays
outage_DN_FD_sim = zeros(1, length(SNR_dB));
outage_DN_FD_ana = zeros(1, length(SNR_dB));

```

```

outage_DF_FD_sim = zeros(1, length(SNR_dB));
outage_DF_FD_ana = zeros(1, length(SNR_dB));

outage_DN_HD_sim = zeros(1, length(SNR_dB));
outage_DN_HD_ana = zeros(1, length(SNR_dB));
outage_DF_HD_sim = zeros(1, length(SNR_dB));
outage_DF_HD_ana = zeros(1, length(SNR_dB));

rate_DN_FD_sim = zeros(1, length(SNR_dB));
rate_DF_FD_sim = zeros(1, length(SNR_dB));
rate_DN_HD_sim = zeros(1, length(SNR_dB));
rate_DF_HD_sim = zeros(1, length(SNR_dB));

%% 3. Main Loop over Transmit SNR
for idx = 1:length(SNR_dB)
    rho = rho_array(idx);

    h2_SN = gamrnd(m, sigma_SN/m, 1, N_sim);
    h2_NF = gamrnd(m, sigma_NF/m, 1, N_sim);
    h2_LI = exprnd(omega_LI, 1, N_sim);

    % FD Calculations
    gamma_DF_to_DN_FD = (h2_SN .* a_F .* rho) ./ ...
        (h2_SN .* a_N .* rho + h2_LI .* rho + 1);
    gamma_DN_FD = (h2_SN .* a_N .* rho) ./ (h2_LI .* rho + 1);
    gamma_2_DF_FD = h2_NF .* rho;

    fail_DN_FD = (gamma_DF_to_DN_FD < gamma_th_F_FD) | ...
        (gamma_DN_FD < gamma_th_N_FD);
    fail_DF_FD = (gamma_DF_to_DN_FD < gamma_th_F_FD) | ...
        ((gamma_DF_to_DN_FD >= gamma_th_F_FD) & ...
        (gamma_2_DF_FD < gamma_th_F_FD));

    outage_DN_FD_sim(idx) = sum(fail_DN_FD) / N_sim;
    outage_DF_FD_sim(idx) = sum(fail_DF_FD) / N_sim;

    rate_DN_FD_sim(idx) = mean(log2(1 + gamma_DN_FD));
    rate_DF_FD_sim(idx) = mean(log2(1 + min(gamma_DF_to_DN_FD, ...
        gamma_2_DF_FD)));

    % HD Calculations
    gamma_DF_to_DN_HD = (h2_SN .* a_F .* rho) ./ ...
        (h2_SN .* a_N .* rho + 1);
    gamma_DN_HD = (h2_SN .* a_N .* rho);
    gamma_2_DF_HD = h2_NF .* rho;

    fail_DN_HD = (gamma_DF_to_DN_HD < gamma_th_F_HD) | ...
        (gamma_DN_HD < gamma_th_N_HD);

```

```

fail_DF_HD = (gamma_DF_to_DN_HD < gamma_th_F_HD) | ...
              ((gamma_DF_to_DN_HD >= gamma_th_F_HD) & ...
               (gamma_2_DF_HD < gamma_th_F_HD));

outage_DN_HD_sim(idx) = sum(fail_DN_HD) / N_sim;
outage_DF_HD_sim(idx) = sum(fail_DF_HD) / N_sim;

rate_DN_HD_sim(idx) = 0.5 * mean(log2(1 + gamma_DN_HD));
rate_DF_HD_sim(idx) = 0.5 * mean(log2(1 + min(gamma_DF_to_DN_HD, ...
                                              gamma_2_DF_HD)));

% Analytical FD
tau_1 = gamma_th_F_FD / (rho * (a_F - a_N * gamma_th_F_FD));
beta_1 = gamma_th_N_FD / (a_N * rho);

if (a_F > a_N * gamma_th_F_FD)
    theta_1 = max(tau_1, beta_1);
    term_DN = (1/omega_LI) * exp(-theta_1/sigma_SN) * ...
              ((1/omega_LI) + (theta_1 * rho / sigma_SN) )^(-1);
    outage_DN_FD_ana(idx) = 1 - term_DN;
    L1 = (1/omega_LI) * exp(-tau_1/sigma_SN) * ...
          ((1/omega_LI) + (tau_1 * rho / sigma_SN) )^(-1);
    outage_DF_FD_ana(idx) = (1 - L1) + ...
                            (1 - exp(-gamma_th_F_FD / (sigma_NF * rho))) * L1;
else
    outage_DN_FD_ana(idx) = 1; outage_DF_FD_ana(idx) = 1;
end

% Analytical HD
tau_2 = gamma_th_F_HD / (rho * (a_F - a_N * gamma_th_F_HD));
beta_2 = gamma_th_N_HD / (a_N * rho);

if (a_F > a_N * gamma_th_F_HD)
    theta_2 = max(tau_2, beta_2);
    outage_DN_HD_ana(idx) = 1 - exp(-theta_2 / sigma_SN);
    outage_DF_HD_ana(idx) = 1 - exp(-(tau_2 / sigma_SN + ...
                                       gamma_th_F_HD / (rho * sigma_NF)));
else
    outage_DN_HD_ana(idx) = 1; outage_DF_HD_ana(idx) = 1;
end

end

%% 4. Throughput
throughput_sys_FD_sim = (1 - outage_DN_FD_sim)*R_N + ...
                        (1 - outage_DF_FD_sim)*R_F;
throughput_sys_FD_ana = (1 - outage_DN_FD_ana)*R_N + ...
                        (1 - outage_DF_FD_ana)*R_F;

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throughput_sys_HD_sim = (1 - outage_DN_HD_sim)*R_N + ...
                        (1 - outage_DF_HD_sim)*R_F;
throughput_sys_HD_ana = (1 - outage_DN_HD_ana)*R_N + ...
                        (1 - outage_DF_HD_ana)*R_F;

%% 5. Sum Ergodic Rate
sum_rate_FD_sim = rate_DN_FD_sim + rate_DF_FD_sim;
sum_rate_HD_sim = rate_DN_HD_sim + rate_DF_HD_sim;

%% 6. Plotting
figure('Name', 'Performance', 'Position', [100, 100, 1500, 450]);

subplot(1,3,1);
semilogy(SNR_dB, outage_DN_FD_sim, 'ro', 'MarkerSize', 5); hold on;
semilogy(SNR_dB, outage_DF_FD_sim, 'rs', 'MarkerSize', 5);
semilogy(SNR_dB, outage_DN_HD_sim, 'bo', 'MarkerSize', 5);
semilogy(SNR_dB, outage_DF_HD_sim, 'bs', 'MarkerSize', 5);
semilogy(SNR_dB, outage_DN_FD_ana, 'r-', 'LineWidth', 1.5);
semilogy(SNR_dB, outage_DF_FD_ana, 'r--', 'LineWidth', 1.5);
semilogy(SNR_dB, outage_DN_HD_ana, 'b-', 'LineWidth', 1.5);
semilogy(SNR_dB, outage_DF_HD_ana, 'b--', 'LineWidth', 1.5);
grid on; ylim([1e-5 1]); xlim([-10 40]);
xlabel('SNR_(dB)'); ylabel('Outage_Probability');
legend('Near_FD', 'Far_FD', 'Near_HD', 'Far_HD', ...
       'Location', 'southwest', 'FontSize', 8);

subplot(1,3,2);
plot(SNR_dB, throughput_sys_FD_sim, 'ro', 'MarkerSize', 5); hold on;
plot(SNR_dB, throughput_sys_HD_sim, 'bs', 'MarkerSize', 5);
plot(SNR_dB, throughput_sys_FD_ana, 'r-', 'LineWidth', 1.5);
plot(SNR_dB, throughput_sys_HD_ana, 'b-', 'LineWidth', 1.5);
grid on; xlim([-10 40]); ylim([0 3.5]);
xlabel('SNR_(dB)'); ylabel('Throughput_(BPCU)');
legend('FD-NOMA', 'HD-NOMA', 'Location', 'northwest', 'FontSize', 8);

subplot(1,3,3);
plot(SNR_dB, rate_DN_FD_sim, 'r^', 'LineWidth', 1.5); hold on;
plot(SNR_dB, rate_DF_FD_sim, 'r-v', 'LineWidth', 1.5);
plot(SNR_dB, sum_rate_FD_sim, 'r-d', 'LineWidth', 1.5);
plot(SNR_dB, rate_DN_HD_sim, 'b--^', 'LineWidth', 1.5);
plot(SNR_dB, rate_DF_HD_sim, 'b--v', 'LineWidth', 1.5);
plot(SNR_dB, sum_rate_HD_sim, 'b--d', 'LineWidth', 1.5);
grid on; xlim([-10 40]);
xlabel('SNR_(dB)'); ylabel('Rate_(BPCU)');
legend('Near_FD', 'Far_FD', 'Sum_FD', 'Near_HD', 'Far_HD', 'Sum_HD', ...
       'Location', 'northwest', 'FontSize', 8);

```


V. REFERENCES

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- [2] M. Bakshi, B. Lall, R. K. Mallik, and A. Singhal, "Performance Analysis of Downlink Cooperative User Relaying FD/HD NOMA Over Nakagami- m Fading Channels," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 8632–8647, June 2024. DOI: 10.1109/TVT.2024.3362429.